

Stage 6 report

Frequency	Theoretical speed	Simulated speed
20 Hz	600 RPM	599.8 RPM
30 Hz	900 RPM	893 RPM
40 Hz	1200 RPM	1198 RPM
50 Hz	1500 RPM	1499 RPM
60 Hz	1800 RPM	1811 RPM mean

Conclusion:

The simulated motor speed followed the expected near-linear relationship between supply frequency and synchronous speed. Small deviations were observed because of transient oscillations and the use of an average-value converter model.

Stage 6

Test 1

Measurement	Result
Frequency	60 Hz
Load step	0 → -20 N·m at 0.5 s
Speed before load	~1820 RPM
Final speed	~1740 RPM
Current before load	27.26 A RMS
Current after load	32.58 A RMS

Test 2A

Measurement	Test 2A result
Frequency command	60 Hz immediately
Load during startup	0 N·m
Peak RMS startup current	108.6 A
Time to ~90% speed (~1620 RPM)	~0.07 s
Peak speed	1953 RPM
Theoretical synchronous speed	1800 RPM
Overshoot	~8.5%
Settled speed before load step	~1800 RPM

Test 2B

Measurement	Direct 60 Hz start	0→60 Hz ramp
Peak RMS current	108.6 A	30.08 A
Time to ~1620 RPM	~0.07 s	0.916 s
Peak speed	1953 RPM	1767 RPM
Final speed	~1800 RPM	~1760 RPM
Speed overshoot	~8.5%	essentially none

Test 3

Measurement	Result
Initial frequency	30 Hz
Final frequency	60 Hz

Frequency step time	0.50 s
Speed before frequency change	~901 RPM
Peak speed after frequency change	~2070 RPM
Final settled speed	~1760–1770 RPM
Approx. settling time after step	~0.30 s
RMS current before frequency change	~7.27 A
Peak RMS current during acceleration	~93.23 A
Final RMS current	~15.13 A

Test 3 observation:

When the VFD frequency command increased from 30 Hz to 60 Hz at 0.5 s, the motor accelerated from approximately 901 RPM to a final speed of about 1760–1770 RPM. The sudden frequency change caused a temporary speed overshoot to approximately 2070 RPM and an RMS current spike of about 93.23 A. After the transient response, the current settled to approximately 15.13 A and the motor speed stabilized near the expected 60 Hz operating region.

Stage 7 Report

Frequency sweep table

Frequency	Theoretical synchronous speed	Simulated speed
20 Hz	600 RPM	599.8 RPM
30 Hz	900 RPM	893 RPM
40 Hz	1200 RPM	1198 RPM
50 Hz	1500 RPM	1499 RPM
60 Hz	1800 RPM	1811 RPM mean

Speed Error

Frequency	Theory	Simulation	Error
20 Hz	600	599.8	-0.03%

30 Hz	900	893	-0.78%
40 Hz	1200	1198	-0.17%
50 Hz	1500	1499	-0.07%
60 Hz	1800	1811	+0.61%

V/f control table

Frequency	Modulation magnitude
20 Hz	0.30
30 Hz	0.45
40 Hz	0.60
50 Hz	0.75
60 Hz	0.90

Load-response results

Measurement	Result
Frequency	60 Hz
Load step	0 $\rightarrow$ -20 N·m
Step time	0.5 s
Speed before load	~1820 RPM
Final speed	~1740 RPM
RMS current before load	27.26 A
RMS current after load	32.58 A

Current increase: $I = 32.58 - 27.26 = 5.32$ A

Percentage increase: $5.32/27.26 * 100 = 19.5\%$

Speed change: $1740 - 1820 = -80 \text{ RPM}$

Percentage speed drop: $80/1820 * 100 = 4.4\%$

Startup comparison

Measurement	Direct 60 Hz start	0→60 Hz ramp
Peak RMS current	108.6 A	30.08 A
Time to ~1620 RPM	~0.07 s	~0.916 s
Peak speed	1953 RPM	1767 RPM
Speed overshoot	~8.5%	essentially none
Final speed	~1800 RPM	~1760 RPM

Peak-current reduction: $108.6 - 30.08 / 108.6 * 100 = 72.3\%$

So there is a clear tradeoff:

Direct start

- faster
- much higher current
- more overshoot

VFD ramp

- slower
- much lower current
- smoother acceleration

Frequency-Step Response

Measurement	Result
Initial frequency	30 Hz
Final frequency	60 Hz
Frequency step time	0.5 s
Speed before change	~901 RPM
Peak speed	~2070 RPM
Final settled speed	~1760–1770 RPM
Approx. settling time	~0.30 s
RMS current before step	~7.27 A
Peak RMS current	~93.23 A
Final RMS current	~15.13 A

$$2070-1800/1800 * 100 = 15\%$$

$$\text{Current increase during acceleration: } 93.23-7.27=85.96\text{A}$$

Graphs:

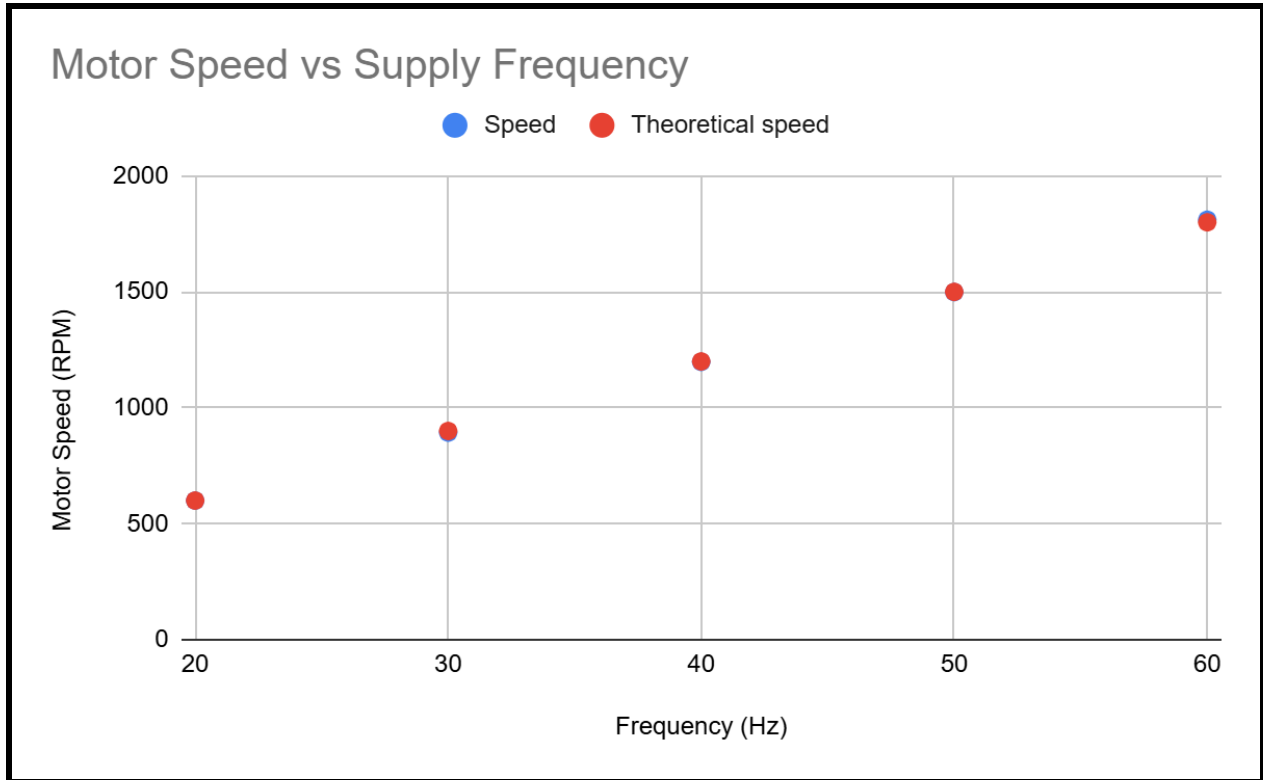


Figure 1. Theoretical and simulated induction motor speed as a function of supply frequency under V/f control. The simulated motor speed closely follows the theoretical synchronous-speed relationship from 20 Hz to 60 Hz.

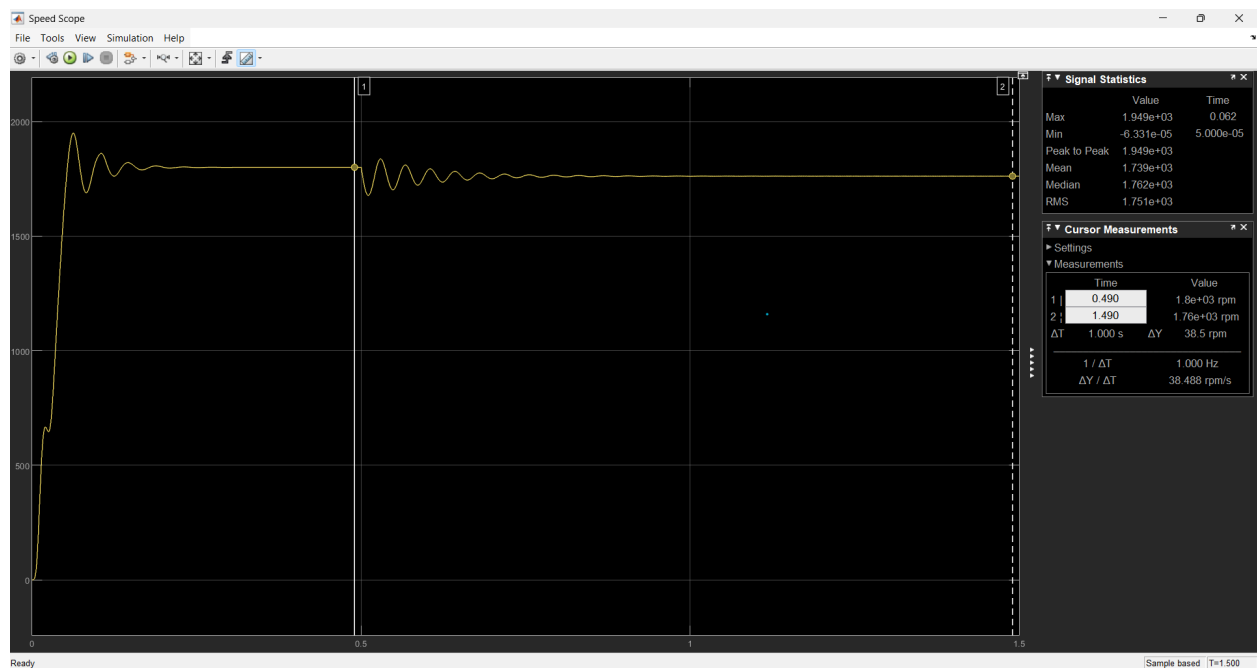


Figure 2. Applying a $-20\text{ N}\cdot\text{m}$ mechanical load reduced motor speed by approximately 80 RPM while RMS current increased from 27.26 A to 32.58 A, an increase of approximately 19.5%. This shows that the motor draws additional current to meet increased torque demand.

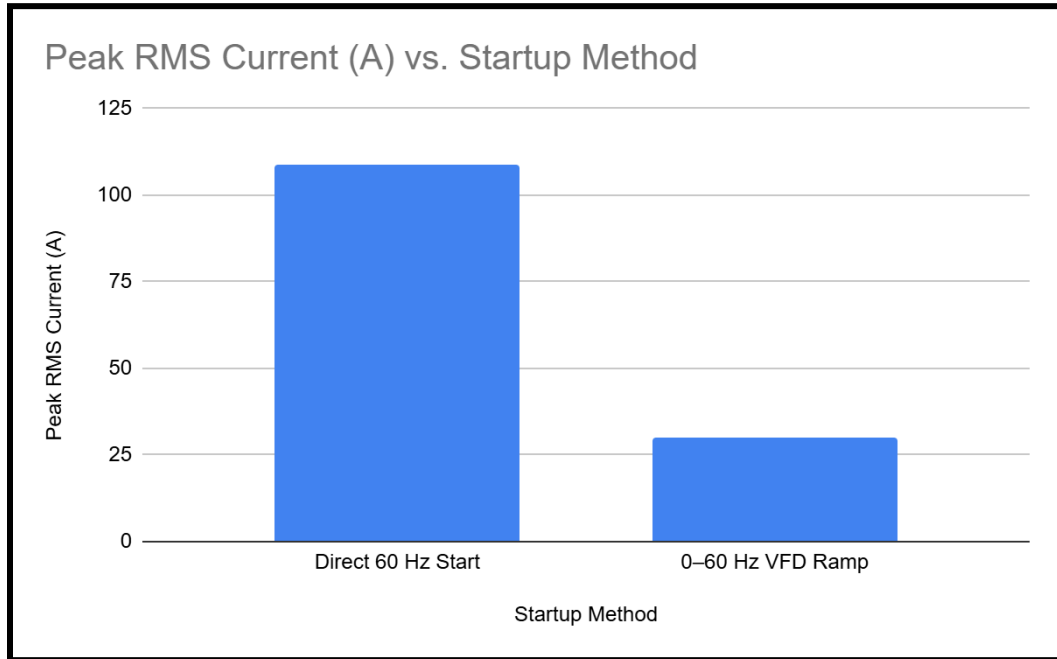


Figure 3. Comparison of peak RMS startup current for a direct 60 Hz start and a 0–60 Hz VFD ramp. The VFD ramp reduced peak startup current from 108.6 A to 30.08 A, a reduction of approximately 72.3%.

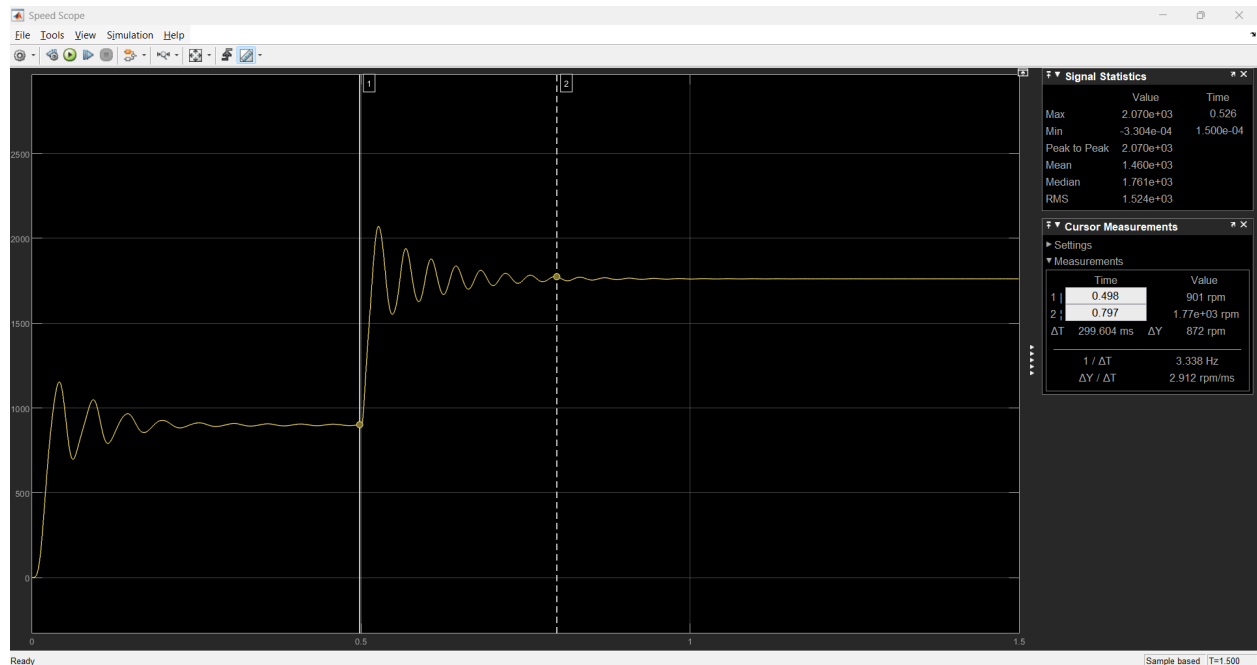


Figure 4. Motor speed response to a 30 Hz to 60 Hz frequency command applied at 0.5 s. The motor accelerates from approximately 900 RPM toward the 60 Hz operating speed, with a transient overshoot before settling.

Conclusion

The simulation demonstrated that open-loop V/f control can regulate the speed of a three-phase induction motor by varying supply frequency while scaling voltage proportionally. Motor speed closely followed the theoretical synchronous-speed relationship from 20 Hz to 60 Hz. Applying mechanical load caused rotor speed to decrease and RMS current to increase. A ramped VFD startup reduced peak RMS current from 108.6 A to 30.08 A, a reduction of approximately 72%, while also reducing speed overshoot. A 30 Hz to 60 Hz frequency step successfully increased motor speed from approximately 901 RPM toward the 60 Hz operating region, although the sudden command produced transient speed overshoot and a large current spike. These results show both the benefits and limitations of open-loop scalar V/f control.